

Ex :

A ^{fair} coin is tossed until a head appears. What is the expectation of the number of tosses required.

Solⁿ.

Let X be the number of tosses required to get the first head. Then

$$E(X) = \sum_{x=1}^{\infty} x p(x)$$

Event	x	$p(x)$
H	1	$\frac{1}{2}$
TH	2	$\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$
TTH	3	$\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$
TTTH	4	$\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{16}$
$\vdots$	$\vdots$	$\vdots$

Hence $E(X) = 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} + 3 \cdot \frac{1}{8} + \dots$

This is the ^{infinite} arithmetic-geometric series

$$S = 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} + 3 \cdot \frac{1}{8} + 4 \cdot \frac{1}{16} + \dots$$

$$\frac{1}{2}S = 1 \cdot \frac{1}{4} + 2 \cdot \frac{1}{8} + 3 \cdot \frac{1}{16} + \dots$$

$$(S - \frac{1}{2}S) = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

$$\frac{1}{2}S = \frac{\frac{1}{2}}{1 - \frac{1}{2}} = 1 \quad (\text{Infinite Geometric Series})$$

$$S = 2, \quad \underline{S = E(X) = 2}$$

Ex. What is the expectation of the number of failures preceding the first success in an infinite series of independent trials with constant prob of success ' p ' in each trial.

Solⁿ. Let x be the number of failures preceding the first success. Then x can take the values $0, 1, 2, \dots$

We have $p(x) = P(X=x) = P[x \text{ failures preced the first success}]$

$$= q^x p, \quad \text{where } q = 1-p \text{ is the prob of failure in a trial}$$

$$\text{then } E(X) = \sum_{x=0}^{\infty} x p(x) = \sum_{x=0}^{\infty} x \cdot q^x p = pq \sum_{x=1}^{\infty} x q^{x-1}$$

$$= pq [1 + 2q + 3q^2 + 4q^3 + \dots]$$

Now $1 + 2q + 3q^2 + 4q^3 + \dots$ is an infinite arithmetic-geometric series

$$\text{Let } S = 1 + 2q + 3q^2 + 4q^3 + \dots$$

$$qS = q + 2q^2 + 3q^3 + \dots$$

$$\therefore (1-q)S = 1 + q + q^2 + q^3 + \dots = \frac{1}{1-q} \quad (\text{infinite geometric series})$$

$$\Rightarrow S = \frac{1}{(1-q)^2}$$

$$\therefore 1 + 2q + 3q^2 + 4q^3 + \dots = \frac{1}{(1-q)^2}$$

$$\text{Hence, } E(X) = \frac{pq}{(1-q)^2} = \frac{pq}{p^2} = \frac{q}{p}$$

$$\boxed{E(X) = \frac{q}{p}}$$

Ex. A box contains 'a' white and 'b' black balls. 'c' balls are drawn. Find the expected value of the number of white balls drawn.

Let a r.v. X_i associated with i th draw, be defined as follows:

$$X_i = \begin{cases} 1 & \text{if } i\text{th ball drawn is white} \\ 0 & \text{if } i\text{th ball drawn is black.} \end{cases}$$

Then the number 'S' of the white balls among 'c' balls drawn is given by

$$S = X_1 + X_2 + \dots + X_c = \sum_{i=1}^c X_i \Rightarrow E(S) = \sum_{i=1}^c E(X_i)$$

$$\text{Now } P(X_i=1) = P(\text{of drawing a white ball}) = \frac{a}{a+b}$$

$$\text{and } P(X_i=0) = P(\text{of drawing a black ball}) = \frac{b}{a+b}$$

$$\therefore E(X_i) = 1 \cdot P(X=1) + 0 \cdot P(X=0) = \frac{a}{a+b}$$

$$\text{Hence } E(S) = \sum_{i=1}^c \left(\frac{a}{a+b} \right) = \frac{ca}{a+b}$$

⇒ Discrete case:

$$\begin{aligned} E[g(X, Y) | Y = y_j] &= \sum_{i=1}^{\infty} g(x_i, y_j) P(X = x_i | Y = y_j) \\ &= \frac{\sum_{i=1}^{\infty} g(x_i, y_j) P(X = x_i \cap Y = y_j)}{P(Y = y_j)} \end{aligned}$$

i.e. $E[g(X, Y) | Y = y_j]$ is nothing but the expectation of the function $g(x, y_j)$ w.r.t. the conditional distribution of X when $Y = y_j$.

In particular: $E(X | Y = y_j) = \sum_{i=1}^{\infty} x_i P(X = x_i | Y = y_j)$

The conditional variance of X given $Y = y_j$ is likewise given by

$$V(X | Y = y_j) = E[\{X - E(X | Y = y_j)\}^2 | Y = y_j]$$

The conditional expectation of $g(X, Y)$ and the conditional variance of Y given $X = x_i$ may also be defined in an exactly similar manner.

⇒ Continuous case: The conditional expectation of $g(X, Y)$ on the hypothesis $Y = y$ is defined by

$$\begin{aligned} E\{g(X, Y) | Y = y\} &= \int_{-\infty}^{\infty} g(x, y) f_{X|Y}(x|y) dx \\ &= \frac{\int_{-\infty}^{\infty} g(x, y) f(x, y) dx}{f_Y(y)} \end{aligned}$$

In particular $E(X | Y = y) = \frac{\int_{-\infty}^{\infty} x f(x, y) dx}{f_Y(y)}$

similarly $E(Y | X = x) = \frac{\int_{-\infty}^{\infty} y f(x, y) dy}{f_X(x)}$

The conditional variance of X may be defined as

$$V(X|Y=y) = E\left[\{X - E(X|Y=y)\}^2 | Y=y\right]$$

Similarly $V(Y|X=x) = E\left[\{Y - E(Y|X=x)\}^2 | X=x\right]$

Ex

The expected value of X is equal to the expectation of the conditional expectation of X given Y .

i.e. $E(X) = E[E(X|Y)]$

Proof: $E[E(X|Y)] = E\left[\sum_i x_i P(X=x_i | Y=y_j)\right]$

$$= E\left[\sum_i x_i \frac{P(X=x_i \cap Y=y_j)}{P(Y=y_j)}\right]$$

$$= \sum_j \left[\sum_i \left\{ x_i \frac{P(X=x_i \cap Y=y_j)}{P(Y=y_j)} \right\} \right] P(Y=y_j)$$

$$= \sum_j \sum_i x_i \cdot P(X=x_i \cap Y=y_j)$$

$$= \sum_i \left[x_i \cdot \left\{ \sum_j P(X=x_i \cap Y=y_j) \right\} \right]$$

$$= \sum_i x_i \cdot P(X=x_i) = E(X)$$

Hence, it is proved.

Ex

Let X and Y be two random variables each taking three values $-1, 0, 1$ and having the following joint prob distⁿ.

$Y \backslash X$	-1	0	1	Total
-1	0	0.1	0.1	0.2
0	0.2	0.2	0.2	0.6
1	0	0.1	0.1	0.2
Total	0.2	0.4	0.4	1.0

- i) Show that X and Y have different expectations
- ii) Prove that X and Y are uncorrelated
- iii) Find variances of X and Y
- iv) Find $P(X|Y=0)$ i.e. conditional prob. distⁿ of X given $Y=0$.
- v) Find $V(Y|X=-1)$

Solⁿ. (i) $E(Y) = \sum y_i p(y_i) = -1(0.2) + 0(0.6) + 1(0.2) = 0$
 $E(X) = \sum x_i p(x_i) = -1(0.2) + 0(0.4) + 1(0.4) = 0.2$
 $\therefore E(X) \neq E(Y)$.

(ii) $E(XY) = \sum_i \sum_j x_i y_j p(x_i y_j) = (-1)(-1)(0) + 0(-1)(0.1) + 1(-1)(0.1)$
 $+ 0(-1)(0.2) + 0(0)(0.2) + 0(1)(0.2)$
 $+ 1(-1)(0) + 1(0)(0.1) + 1(1)(0.1)$
 $= 0$

$$\text{Cov}(XY) = E(XY) - E(X)E(Y) = 0$$

$\Rightarrow X$ & Y are uncorrelated.

iii) $E(Y^2) = (-1)^2(0.2) + 0^2(0.6) + 1^2(0.2) = 0.4$

$$V(Y) = E(Y^2) - (E(Y))^2 = 0.4$$

$$E(X^2) = (-1)^2(0.2) + 0^2(0.4) + 1^2(0.4) = 0.6$$

$$V(X) = E(X^2) - (E(X))^2 = 0.56$$

iv) $P(X=-1|Y=0) = \frac{P(X=-1, Y=0)}{P(Y=0)} = \frac{0.2}{0.6} = \frac{1}{3}$

$$P(X=0|Y=0) = \frac{P(X=0, Y=0)}{P(Y=0)} = \frac{0.2}{0.6} = \frac{1}{3}$$

$$P(X=1|Y=0) = \frac{P(X=1, Y=0)}{P(Y=0)} = \frac{0.2}{0.6} = \frac{1}{3}$$

v) $V(Y|X=-1) = E(Y|X=-1)^2 - \{E(Y|X=-1)\}^2$

$$E(Y|X=-1) = \sum y p(Y=y|X=-1) = (-1)(0) + 0(0.2) + 1(0) = 0$$

$$E(Y|X=-1)^2 = \sum y^2 p(Y=y|X=-1) = 1(0) + 0(0.2) + 1(0) = 0$$

$$\therefore V(Y|X=-1) = 0$$

Ex. $f(x_1, x_2) = 21x_1^2x_2^3$, $0 < x_1 < x_2 < 1$ be the joint prob distⁿ of X_1 and X_2 . Find the conditional mean and variance of X_1 given $X_2 = x_2$, $0 < x_2 < 1$.

Solⁿ

Marginal of X_2 is given by

$$f_2(x_2) = \int_0^{x_2} f(x_1, x_2) dx_1 = 21x_2^3 \int_0^{x_2} x_1^2 dx_1 = 21x_2^3 \left[\frac{x_1^3}{3} \right]_0^{x_2} \\ = 7x_2^6; \quad 0 < x_2 < 1$$

Conditional pdf of X_1 (given X_2) is given by

$$f_1(x_1|x_2) = \frac{f(x_1, x_2)}{f_2(x_2)} = 3 \cdot \frac{x_1^2}{x_2^3}; \quad 0 < x_1 < x_2; \quad 0 < x_2 < 1$$

Conditional mean of X_1 is

$$E(X_1|X_2=x_2) = \int_0^{x_2} x_1 f_1(x_1|x_2) dx_1 = \frac{3}{x_2^3} \int_0^{x_2} x_1^3 dx_1 \\ = \frac{3}{x_2^3} \left[\frac{x_1^4}{4} \right]_0^{x_2} = \frac{3x_2}{4}, \quad 0 < x_2 < 1.$$

$$\text{Now } E(X_1^2|X_2=x_2) = \int_0^{x_2} x_1^2 f_1(x_1|x_2) dx_1 = \frac{3}{x_2^3} \int_0^{x_2} x_1^4 dx_1 \\ = \frac{3}{x_2^3} \left[\frac{x_1^5}{5} \right]_0^{x_2} = \frac{3}{5} x_2^2$$

$$\therefore \text{Var}(X_1|X_2=x_2) = E(X_1^2|X_2=x_2) - (E(X_1|X_2=x_2))^2 \\ = \frac{3}{5} x_2^2 - \frac{9}{16} x_2^2 = \frac{3}{80} x_2^2; \quad 0 < x_2 < 1.$$

Ex. Two random variables X and Y have the following joint prob density function:

$$f(x, y) = \begin{cases} 2^{-x-y} & ; 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

- i) Find Marginal prob density function of X and Y .
- ii) Conditional density functions (iii) $\text{Var}(X)$ and $\text{Var}(Y)$
- (iv) $\text{Cov}(X, Y)$

Solⁿs :

$$1) f_X(x) = \int_{-\infty}^{\infty} f(x,y) dy = \int_0^1 (2-x-y) dy = \frac{3}{2} - x$$

$$\text{Hence } f_X(x) = \begin{cases} \frac{3}{2} - x, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Similarly } f_Y(y) = \begin{cases} \frac{3}{2} - y, & 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

$$(II) f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)} = \frac{(2-x-y)}{(\frac{3}{2}-y)} ; 0 < (x,y) < 1$$

$$f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)} = \frac{(2-x-y)}{(\frac{3}{2}-x)} ; 0 < (x,y) < 1$$

$$E(X) = \int_0^1 x f_X(x) dx = \int_0^1 x (\frac{3}{2} - x) dx = \frac{5}{12}$$

$$E(Y) = \int_0^1 y f_Y(y) dy = \int_0^1 y (\frac{3}{2} - y) dy = \frac{5}{12}$$

$$E(X^2) = \int_0^1 x^2 f_X(x) dx = \int_0^1 x^2 (\frac{3}{2} - x) dx = \frac{1}{4}$$

$$V(X) = E(X^2) - (E(X))^2 = \frac{1}{4} - \frac{25}{144} = \frac{11}{144}$$

$$\text{Similarly } V(Y) = \frac{11}{144}$$

$$iv) \text{Cov}(X,Y) = E(XY) - E(X)E(Y)$$

$$E(XY) = \int_0^1 \int_0^1 xy (2-x-y) dx dy = \frac{1}{6}$$

$$\therefore \text{Cov}(X,Y) = \frac{1}{6} - \frac{5}{12} \cdot \frac{5}{12} = -\frac{1}{144}$$

Ex.

$$f(x,y) = 8xy, \quad 0 < x < y < 1 \\ = 0, \quad \text{elsewhere}$$

Find (a) $E(Y|X=x)$ (b) $E(XY|X=x)$ (c) $\text{Var}(Y|X=x)$

$$\text{Solⁿ. } f_X(x) = \int_{-\infty}^{\infty} f(x,y) dy = 8x \int_x^1 y dy = 4x(1-x^2), \quad 0 < x < 1$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y) dx = 8y \int_0^y x dx = 4y^3, \quad 0 < y < 1$$

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{2x}{y^2},$$

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{2y}{1-x^2}, \quad 0 < x < y < 1$$

$$(a) \quad E(Y|X=x) = \int_0^1 y \left(\frac{2y}{1-x^2} \right) dy = \frac{2}{3} \frac{(1-x^3)}{(1-x^2)} = \frac{2}{3} \frac{(1+x+x^2)}{(1+x)}$$

$$(b) \quad E(XY|X=x) = x E(Y|X=x) = \frac{2}{3} \frac{x(1+x+x^2)}{(1+x)}$$

$$(c) \quad E(Y^2|X=x) = \int_0^1 y^2 \left(\frac{2y}{1-x^2} \right) dy = \frac{1}{2} \frac{(1-x^4)}{(1-x^2)} = \frac{1+x^2}{2}$$

$$\begin{aligned} \text{Var}(Y|X=x) &= E(Y^2|X=x) - (E(Y|X=x))^2 \\ &= \frac{1+x^2}{2} - \frac{4}{9} \frac{(1+x+x^2)^2}{(1+x)^2} \end{aligned}$$

Ex. $f(x) = 6x(1-x); \quad 0 \leq x \leq 1$

i) Check $f(x)$ is pdf ii) obtain DF (or CDF) of X

iii) Compute $P(X \leq \frac{1}{2} | \frac{1}{3} \leq X \leq \frac{2}{3})$ (iv) Determine K such that $P(X < K) = P(X > K)$

i) $f \sim \text{pdf} \quad \int_0^1 f(x) dx = \int_0^1 6x(1-x) dx = 1$

ii) $F_X(x) = P(X \leq x) = \int_0^x 6x(1-x) dx$
 $= (3x^2 - 2x^3); \quad 0 \leq x \leq 1$

iii) $P(X \leq \frac{1}{2} | \frac{1}{3} \leq X \leq \frac{2}{3}) = \frac{P(\frac{1}{3} \leq X \leq \frac{1}{2})}{P(\frac{1}{3} \leq X \leq \frac{2}{3})} = \frac{\int_{\frac{1}{3}}^{\frac{1}{2}} 6x(1-x) dx}{\int_{\frac{1}{3}}^{\frac{2}{3}} 6x(1-x) dx}$
 $= \frac{\frac{11}{26}}{\frac{1}{3}} = \frac{11}{26}$

iv) $P(X < K) = P(X > K)$
 $= \int_0^K 6x(1-x) dx = \int_K^1 6x(1-x) dx \quad \Rightarrow 3K^2 - 2K^3 = 3(1-K^2) - 2(1-K^3)$
 $\Rightarrow 4K^3 - 6K^2 + 1 = 0 \Rightarrow K = \frac{1}{2}, \frac{1+\sqrt{3}}{2}$
 The only admissible value in the given range is $\frac{1}{2}$.
 Hence $K = \frac{1}{2}$.